

ASTRAKAVACH

Autonomous **Find - Patch - Prove** for Indian Defence Software

01 THE THREAT

Indian defence software faces persistent, targeted cyber threats — including campaigns associated with **APT36 / Transparent Tribe**.

02 THE GAP

Scanners find but don't fix.
Manual patch cycles are slow.
LLM-only fixes can be unverified.

03 OUR IDEA

A closed-loop **cyber-reasoning system** combining fuzzing, static/dynamic analysis, deterministic patching, and an **offline LLM as a controlled patch assistant**.

04 WHAT IT DOES

Finds → patches → proves vulnerabilities through an automated verification loop.

TEAMMATES

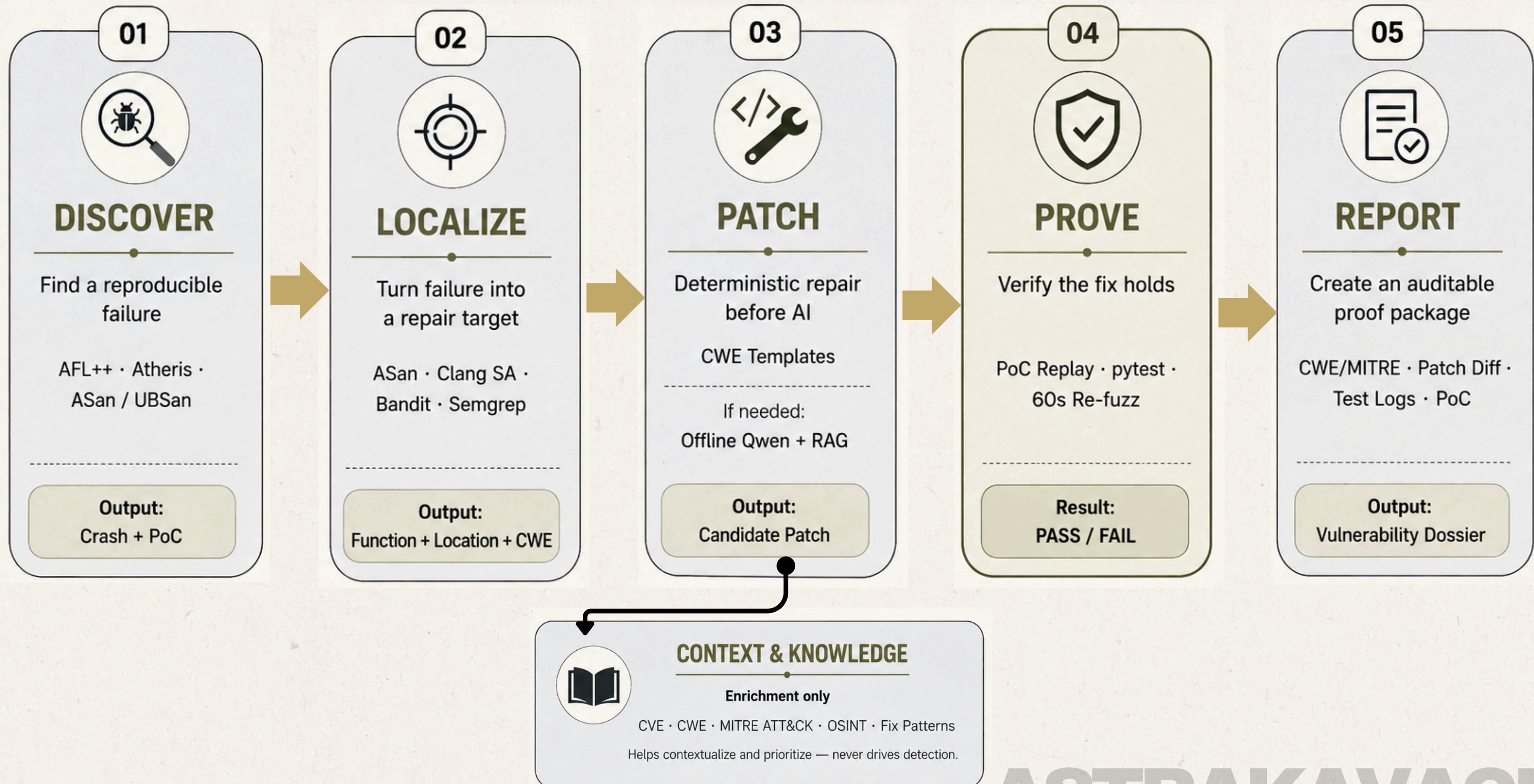
NEERAJ GANESH PAREKH
ELECTRONICS 3RD YEAR

NISHANT KUMAR
CSE-AIML 2ND YEAR

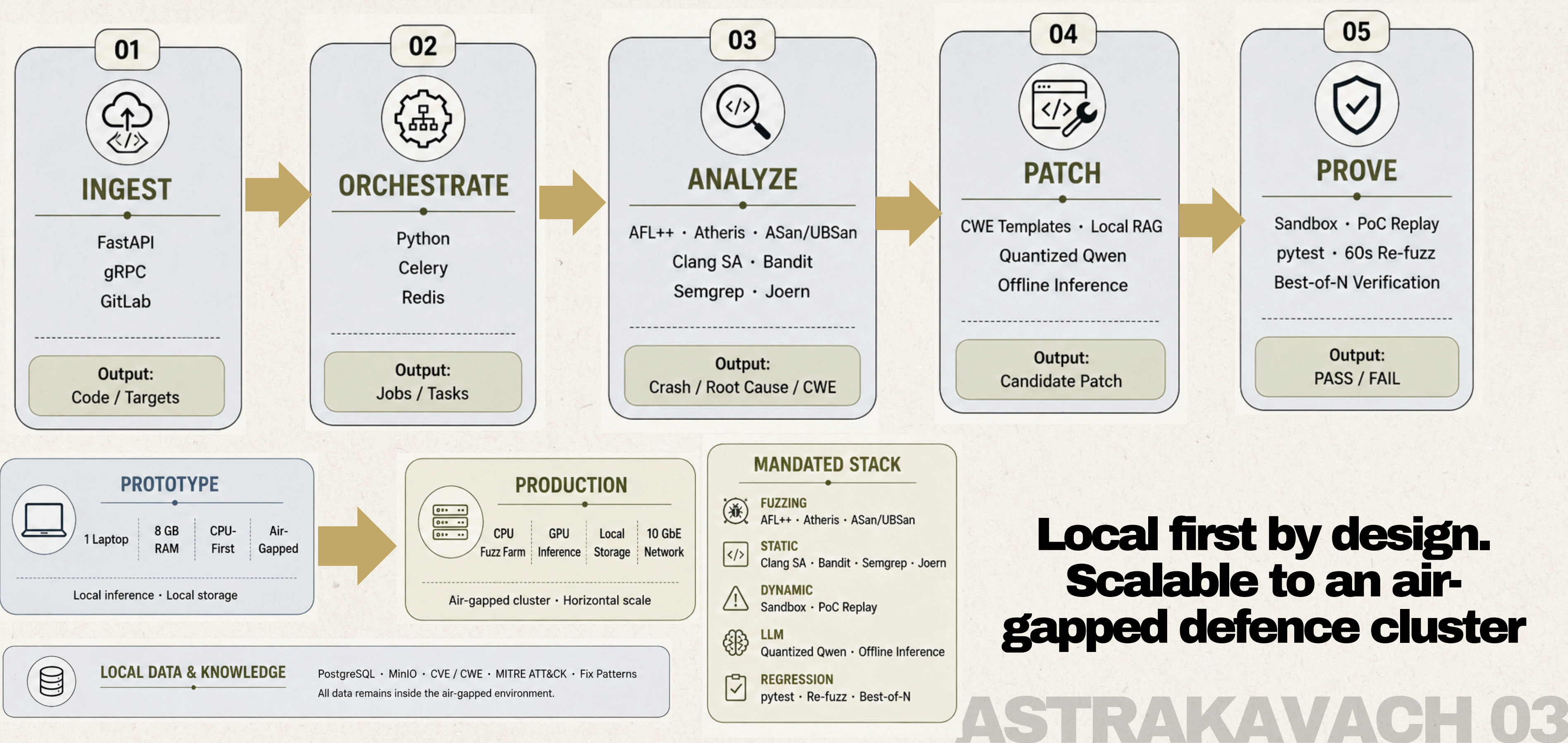
ASHUTOSH SHRI
2ND YEAR

DETAILED METHODOLOGY

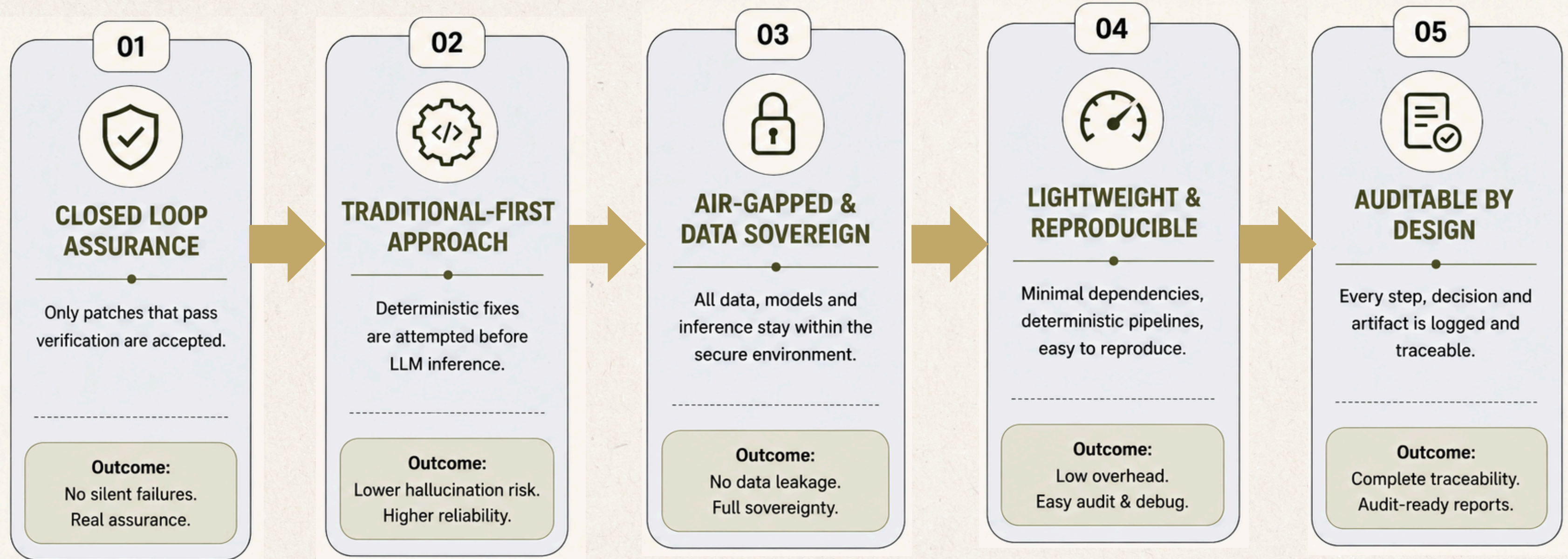
Traditional-first, LLM-last: a closed loop that proves itself



TECHNOLOGY STACK & ARCHITECTURE



SALIENT FEATURES & NOVELTY



KEY BENEFITS

- Autonomous discovery to verified patch.
- Reduced mean time to remediation.
- Fewer false positives and patch churn.
- Defence-grade trust and traceability.
- Designed for real-world constraints.

COMPARED TO TRADITIONAL TOOLS

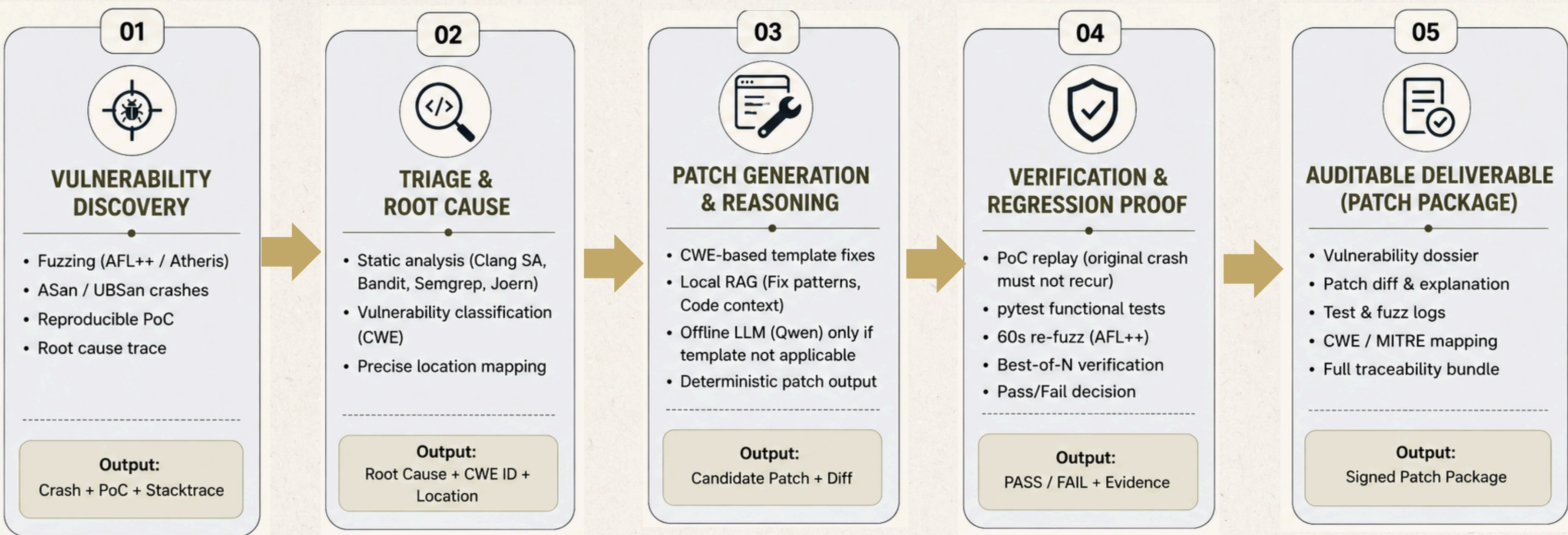
Traditional Tools	AstraKavach
Find vulnerabilities	Find + fix + prove
Manual triage & fix	Automated reasoning
No proof of fix	Verified by harness
High false positives	Low false positives
Siloed outputs	Unified proof package

DEFENCE-GRADE IMPACT

- Stronger security posture.
- Compliance and audit readiness.
- Supports secure SDLC adoption.
- Scales from workstation to cluster.
- Built for Indian defence ecosystems.

Why AstraKavach is lightweight, verified, and defence-ready.

FINAL DELIVERABLES & POC



DELIVERABLES (WHAT WE SUBMIT)

- ✓ AstraKavach core framework (offline, reproducible)
- ✓ Sample vulnerability → patch → proof walkthrough
- ✓ Patch package examples (diff, logs, dossier)
- ✓ Test results & verification reports
- ✓ Architecture & deployment guide
- ✓ User & operator documentation (air-gapped setup)
- ✓ Repository with reproducible instructions & datasets

SUCCESS METRICS

🎯 Detection Coverage	Target > 90% (fuzz + static)
🛡️ Patch Acceptance Rate	Target > 80%
🕒 Mean Time to Remediate	Target < 4 hours
🐛 False Positive Rate	Target < 10%
🔄 Re-fuzz Validation	Target 100% crash-free

STAKEHOLDERS & USERS

- Defence Developers & Engineers
- Security Analysts & Red Teams
- Quality Assurance Teams
- Programme Managers
- CERT-In / Defence CERTs
- Audit & Compliance Authorities

**From vulnerability to proof:
what we build, measure, and
demonstrate.**

END STATE | AstraKavach continuously discovers, patches and proves vulnerabilities — delivering verified, auditable and deployable patches that strengthen the security posture of Indian defence software systems.

